

LPV Gain-Scheduling Control for an Inverted Pendulum on a Cart

1 Introduction

This document outlines an LPV (Linear Parameter-Varying) gain-scheduling control design for an inverted pendulum system on a cart. The approach involves computing a set of LQR gains at different operating points defined by the pendulum angle deviation and then interpolating these gains in real time. This adaptive strategy ensures that the controller maintains desirable performance despite variations in the system dynamics due to changes in the operating point.

2 LPV Gain-Scheduling Design

2.1 Nominal State-Space Model

The plant is modeled by a state-space representation:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{bd}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{ad}{\Delta} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{c}{\Delta} \\ 0 \\ -\frac{b}{\Delta} \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The common terms are defined as:

$$a = 2m_w + m_p + \frac{2I_w}{r^2}, \quad b = m_p l, \quad c = m_p l^2 + I_p,$$

$$d = m_p g l, \quad \Delta = ac - b^2.$$

This nominal model is defined at a scheduling parameter $p = 0$.

2.2 Operating Point Variation via Scheduling Parameter

In the LPV framework, the scheduling parameter p is chosen as the pendulum angle deviation. For small deviations, the gravitational effect on the system can be approximated by:

$$d_{\text{eff}} = m_p g l \cos(p).$$

The only modification in the state-space matrix is assumed to be in the $(4, 3)$ element of A :

$$A_{\text{lpv}}(4, 3) = \frac{a d_{\text{eff}}}{\Delta}.$$

A range for p is defined (e.g., $-0.1 \leq p \leq 0.1$ rad), and LQR gains are computed at a finite number of operating points over this range.

2.3 LQR Gain Computation

At each operating point p_i in the range, the LQR design is performed with weighting matrices:

$$Q = \text{diag}([100, 1, 1000, 1]), \quad R = 0.01.$$

For each p_i , the modified A matrix is:

$$A_{\text{lpv}} = A, \quad \text{with } A_{\text{lpv}}(4, 3) = \frac{a (m_p g l \cos(p_i))}{\Delta}.$$

Then the LQR gain $K(p_i)$ is computed:

$$K(p_i) = \text{lqr}(A_{\text{lpv}}, B, Q, R).$$

These gains are stored and later used for interpolation.

3 Gain Interpolation and Controller Implementation

An interpolation function is defined:

$$K_{\text{interp}}(p) = \text{interp1}(p_range, \{K(p_i)\}, p, \text{'linear'}, \text{'extrap'}).$$

At each time step during simulation, the current scheduling parameter p_{current} is taken as the current pendulum angle x_3 . The interpolated gain K_{current} is then applied in the state feedback control law:

$$u = -K_{\text{current}} x.$$

This real-time gain adjustment allows the controller to adapt to the nonlinear variation in the plant dynamics.

4 Closed-Loop Simulation

The closed-loop dynamics are simulated using the nominal state-space model (A, B) . The simulation procedure involves:

1. Initializing the state vector $x(0) = [0; 0; 0.1; 0]$ (with an initial pendulum angle deviation of 0.1 rad).

2. At each time step, extracting the current p_{current} from x_3 .
3. Interpolating the gain $K_{\text{current}} = K_{\text{interp}}(p_{\text{current}})$.
4. Computing the control input $u = -K_{\text{current}} x$.
5. Updating the state using Euler integration:

$$x(k+1) = x(k) + dt \left(A x(k) + B u(k) \right).$$

5 Tuning Parameters and Their Effects

Key parameters and their effects on performance include:

- **Weighting Matrices Q and R :** These determine the baseline LQR performance. Increasing the weight on the cart position or pendulum angle leads to more aggressive control but may result in higher control effort.
- **Scheduling Parameter Range:** A wider range captures more dynamic variations but requires careful interpolation to avoid abrupt changes in control gains.
- **Interpolation Method:** Linear interpolation is used here; alternative interpolation techniques (such as spline interpolation) may offer smoother transitions.

By fine-tuning these parameters, the LPV controller is able to effectively adapt to changes in operating conditions and maintain stability and performance.

6 Simulation Results and Analysis

The simulation results are typically presented via plots:

- **Cart Position:** A plot of $x_1(t)$ shows the tracking performance for the cart.
- **Pendulum Angle:** A plot of $x_3(t)$ indicates how the pendulum angle is regulated.
- **Control Input:** The control input $u(t)$ is monitored to ensure it remains within practical bounds.

These results help assess the transient response and steady-state performance of the LPV gain-scheduling controller.

7 Summary

The LPV gain-scheduling approach provides an effective strategy for controlling nonlinear systems by adapting the control gains based on a scheduling parameter—in this case, the pendulum angle deviation. The design process involves:

- Computing a family of LQR gains over a range of operating points.
- Interpolating these gains in real time based on the current state.
- Implementing a closed-loop control law that updates the control input at each time step.

This method allows the controller to accommodate variations in the system dynamics and maintain robust performance across the operating envelope.